

Hamiltonian Neural Networks for Solar System

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Abstract

This paper investigates a structurally constrained, physics-informed geometric deep learning framework designed to model a 10-body configuration of the Solar System over astronomical timescales up to 400 years. Conventional Hamiltonian Neural Networks (HNNs) frequently exhibit optimization stagnation and localized numerical instabilities—typically leading to unphysical inner planet ejections within 10 to 50 years—due to the high dynamic range of physical magnitudes across celestial bodies. To address these baseline characteristics, we employ a Vector-wise Multiscale V2 normalization scheme alongside an explicit Kinetic-Potential (T-V) energy separation via the Plausible HNN (P-HNN). While dense P-HNN architectures extend the sustainable orbital lifespan up to 200 years, scaling the framework over a 4,000-year dataset to encompass Pluto's full orbital threshold (~ 248 years) introduces parameter inflation and computational bottlenecks during training. To manage this overhead, we develop the Block (Sparse) Plausible HNN (BLK-P-HNN), which incorporates a Topology Adaptor to isolate position inputs into 45 independent sub-manifolds via batch matrix multiplications. This structural sparsity compresses the model's total weight footprint 400-fold, from 88.6 MB down to 200 kB. Empirical evaluation across the 400-year integration window demonstrates a distinct cost-performance trade-off: while the dense P-HNN maintains a lower average tracking error and isotropic dispersion, the compressed BLK-P-HNN successfully maintains a gravitationally bound state for the entire 400-year duration, albeit at the expense of introducing subtle localized orbital variations and increased dispersion along the Z-axis. Ultimately, these results demonstrate that embedding domain-specific inductive biases with structural sparsity provides a viable pathway to achieve extreme parameter compression while preserving macro-scale gravitational stability in long-term multi-body celestial simulations.

1. Introduction

The long-term orbital prediction of multi-body gravitational configurations remains a foundational yet computationally formidable challenge in celestial mechanics [1, 2]. Historically, deterministic

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trajectories of complex stellar and planetary systems have been mapped utilizing high-order symplectic integrators, such as the Wisdom-Holman or Yoshida schemes, to preserve the structural geometric invariants of Hamiltonian systems [3]. While these classical numerical algorithms deliver exceptional fidelity, integrating chaotic or high-dimensional systems over astronomical timescales demands an intractable computational burden, rendering rapid, iterative long-term simulations prohibitively expensive [4]. To alleviate this operational overhead, the recent paradigm of Physics-Informed Neural Networks (PINNs) has emerged as a promising alternative, seeking to leverage the universal function approximation capabilities of deep learning to bypass expensive step-by-step numerical integration [5].

A pivotal milestone in this domain was the inception of Hamiltonian Neural Networks (HNNs) by Greydanus et al. (2019), which mathematically enforce the conservation laws of symplectic geometry by embedding Hamilton’s equations directly into the network’s optimization objective [6]. Despite their theoretical elegance in preserving total energy, we identify that these conventional black-box HNN architectures have primarily been validated only on low-dimensional, isolated toy problems—such as 2-body ideal pendulums or simple planetary pairs—and fail catastrophically when scaled to realistic multi-body scenarios [7, 8]. When attempting to model a highly coupled 10-body configuration of the Solar System, standard dense HNN architectures encounter a severe High Dynamic Range (HDR) problem, characterized by extreme disparities in physical magnitudes (mass, distance, and orbital velocity) across inner and outer celestial bodies. Under these unmitigated HDR conditions, standard models inevitably succumb to localized geometric instabilities and cascading energetic failures, culminating in a domino effect of unphysical inner planetary ejections within short integration windows of merely 10 to 50 years [9, 10].

To systematically break this geometric barrier and mitigate multi-body trajectory drifts, this paper introduces, for the first time, the Plausible HNN (P-HNN) framework alongside its structurally optimized, next-generation evolution: the Block (Sparse) Plausible Hamiltonian Neural Network (BLK-P-HNN). Shifting entirely away from the generic black-box topologies used in standard HNNs, our newly proposed P-HNN architecture enforces strict structural inductive biases by executing an explicit Kinetic-Potential (T-V) energy separation and exploiting functional spatial symmetries via even activation functions. Combined with our novel Vector-wise Multiscale V2 normalization standard specifically tailored to handle the Solar System’s HDR landscape, this baseline dense P-HNN successfully extends the system-level lifespan up to 200 years within unseen test domains, representing an immediate advancement over conventional HNN boundaries by preventing catastrophic orbital decay.

To manage these architectural constraints and investigate the balance between model efficiency and long-term macro-scale stability, we develop the BLK-P-HNN architecture. The defining structural element introduced in this framework is a Topology Adaptor that isolates position inputs into 45 independent planet-to-planet sub-manifolds via batch matrix multiplications. By systematically eliminating redundant cross-planet weight connections that do not correspond to physical pairwise interactions, the BLK-P-HNN implements a structured sparsity that reduces the model’s total weight footprint 400-fold—from 88.6 MB down to 200 kB. Empirical tracking over a 400-year integration window reveals an informative performance trade-off: while the extreme parameter compression introduces subtle localized orbital variations and an increased error dispersion concentrated along the vertical Z-axis compared to the dense baseline, the sparse network successfully maintains a gravitationally bound state and prevents catastrophic multi-body planetary ejections throughout the entire astronomical horizon.

In summary, the primary contributions of this work are twofold:

1. We analyze the scalability boundaries of standard HNNs under high-dynamic-range constraints, and demonstrate through the proposed P-HNN framework how explicit Kinetic-Potential (T-V) separation serves to suppress catastrophic multi-body orbital decay.
2. We evaluate the BLK-P-HNN architecture governed by a batch-matrix-multiplication Topology Adaptor, providing a quantitative mapping of the trade-off between extreme parameter compression (400-fold reduction) and long-term macro-scale gravitational bound-state stability over a 400-year horizon.

2. Data Preparation

The dataset have been synthetically generated by using the 12th order Yoshida symplectic integrator with NASA data on 00h:00m:00s on January 1st, 2026 [16].

3. Methodology

Section 3.1: Conventional HNN Architecture (Baseline)

The conventional HNN, denoted as the baseline model, follows the canonical design proposed by Greydanus et al. (2019). The architecture is designed to map a 60-dimensional state vector—comprised of scaled positions $\hat{q}$ and momenta $\hat{p}$ for the 10 celestial bodies—to a single scalar value representing the Hamiltonian in the scaled state space, $H(\hat{q}, \hat{p})$.

Structural Specification:

- **Input Layer:** A 60-element tensor $(\hat{q}_0, \dots, \hat{q}_{29}; \hat{p}_0, \dots, \hat{p}_{29})$.
- **Hidden Layers:** Two consecutive blocks of fully connected layers (nn.Linear) followed by the hyperbolic tangent (nn.Tanh) activation function. All linear layers include learnable **bias** terms by default.
- **Output Layer:** A final linear layer reducing the hidden features to a 1-dimensional scalar H .

Learning Mechanism:

The model is trained by embedding the symplectic structure directly into the $(\hat{q}, \hat{p})$ state space. The network does not learn the coordinates directly; instead, it optimizes the Hamiltonian surface such that its partial derivatives satisfy the Hamilton's equations:

$$\frac{dp}{dt} = -\frac{\partial H}{\partial q}, \quad \frac{dq}{dt} = \frac{\partial H}{\partial p}$$

By backpropagating through these gradients, the system ensures that the phase space flow remains on the manifold of constant energy, at least locally within the scaled domain.

Section 3.2: Plausible HNN Architecture (Bridge Model)

To address the limitations of the black-box approach, we developed the Plausible HNN, an intermediate architecture designed to validate the impact of explicit energy separation and functional symmetry. This model serves as a crucial baseline for demonstrating the effectiveness of the Even Activation function before transitioning to the sparse block-based design.

1. Kinetic-Potential Energy Separation (T-V Split)

Unlike the conventional HNN, which uses a single MLP to map the entire state vector to a scalar, the Plausible HNN strictly enforces the additive nature of the Hamiltonian: $H = T(\hat{p}) + V(\hat{q})$.

- **T-net (Analytical Prior):** The kinetic energy is modeled as a weighted quadratic sum of momenta, $T = \sum w \cdot \hat{p}^2$. By using a learnable parameter w for each dimension, the network captures the diagonal components of the kinetic energy matrix directly.
- **V-net (Dense MLP):** The potential energy is modeled using a Dense MLP that receives only the position vector $\hat{q}$.

2. Symmetry Bias via Even Activation

The defining feature of this model is the introduction of a custom Even Activation function:

$$f(x) = \ln(1 + x \cdot \tanh(x))$$

This function is strictly symmetric ($f(x) = f(-x)$), ensuring that the learned potential field reflects the even parity often required for stable energy surfaces. This architectural bias prevents the model from learning unphysical odd-order gradients that often cause the rapid orbital collapse (divergence) observed in standard Tanh-based HNNs.

Section 3.3: Block (Sparse) Plausible HNN Architecture

To achieve extreme parametric efficiency, we implemented a Block Sparse connectivity pattern that decomposes the global potential energy into independent pairwise contributions. The architecture is realized through the following specialized modules:

1. Pairwise Input Linear Adaptor (Data Routing)

The initial layer, Topology Adaptor, acts as a physical router. Instead of processing the entire 30-dimensional position tensor $\hat{q}$ globally, it utilizes a pre-computed indexing buffer to extract 45 unique celestial pairs (${}_{10}C_2$).

- **Mechanism:** For each pair (i, j) , it slices a 6-dimensional sub-tensor representing the 3D coordinates of both bodies.

- **Parallel Processing:** These pairs are processed in parallel using Batch Matrix Multiplication (BMM), ensuring that the initial linear transformation is applied independently to each interaction block.

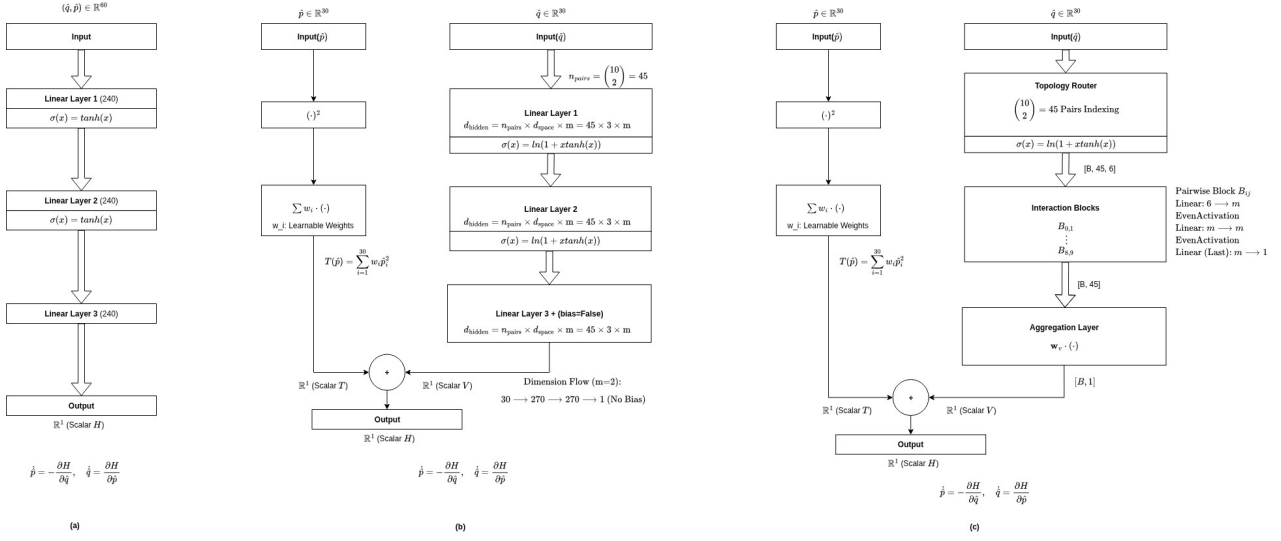


Figure 1: Architectural evolution of Hamiltonian Neural Networks (HNNs) for celestial dynamics. (a) Conventional HNN mapping the global 60-dimensional scaled state vector via a standard dense black-box MLP; (b) Plausible HNN enforcing analytical prior via explicit Kinetic-Potential (T-V) energy separation and parity symmetry through custom Even Activation; and (c) Block Plausible HNN hardcoding the pairwise topology to maximize parametric efficiency.

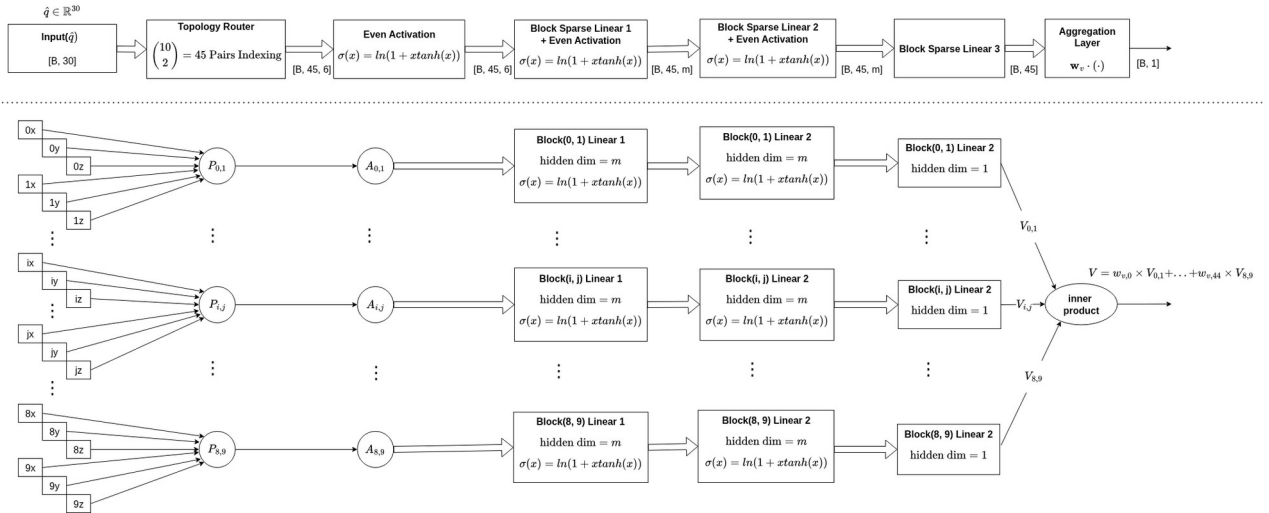


Figure 2. Detailed schematic of the Pairwise Topology Router and data slicing mechanism within the Block Plausible HNN. The global position tensor $X \in \mathbb{R}^{B \times 30}$ is explicitly mapped into $n_{\text{pairs}} = {}_{10}C_2 = 45$ isolated interaction sub-manifolds ($\mathbb{R}^{B \times 1 \times 6}$) using a static indexing buffer. Each block extracts unique coordinates (e.g., $(0_x, 0_y, 0_z)$ and $(1_x, 1_y, 1_z)$) for $\text{Pair}_{0,1}$ to enforce independent, parallel feature extraction via Batch Matrix Multiplication (BMM).

2. Pairwise Hidden Block Layers (Modular Evolution)

The intermediate layers consist of PairWiseHiddenBlockLinear modules. Unlike conventional dense layers where every neuron connects to every neuron in the next layer, this module maintains 45 isolated weight manifolds.

- **Sparsity:** Information regarding the interaction of Earth-Mars, for instance, never mixes with Sun-Jupiter data within the hidden layers.
- **Efficiency:** This structural isolation reduces the parameter count from a $O(N^2)$ dense matrix to a block-diagonal-like structure, directly contributing to the reduction of the model size to $\sim$ hundreds of kilo-Bytes.

3. Attention-based Potential Aggregation

The final scalar potential V is not a simple sum but a weighted aggregation.

- **Weighted Summation:** The network produces 45 individual scalar potentials V_{ij} . These are then combined via a learnable weight vector $\mathbf{w}_v$ (Attention-like mechanism): $V = \mathbf{w}_v^T \mathbf{V}_{\text{pair}}$.
- **Physical Significance:** This allows the model to learn the relative significance (e.g., gravitational coupling strength) of each pair.

The implementation of these novel HNN architectures, Plausible HNN (P-HNN) and Block Sparse Plausible HNN (BLK-P-HNN) are available on the github [17].

4. Results and Discussions

Section 4.1: Limitations of Baseline HNNs

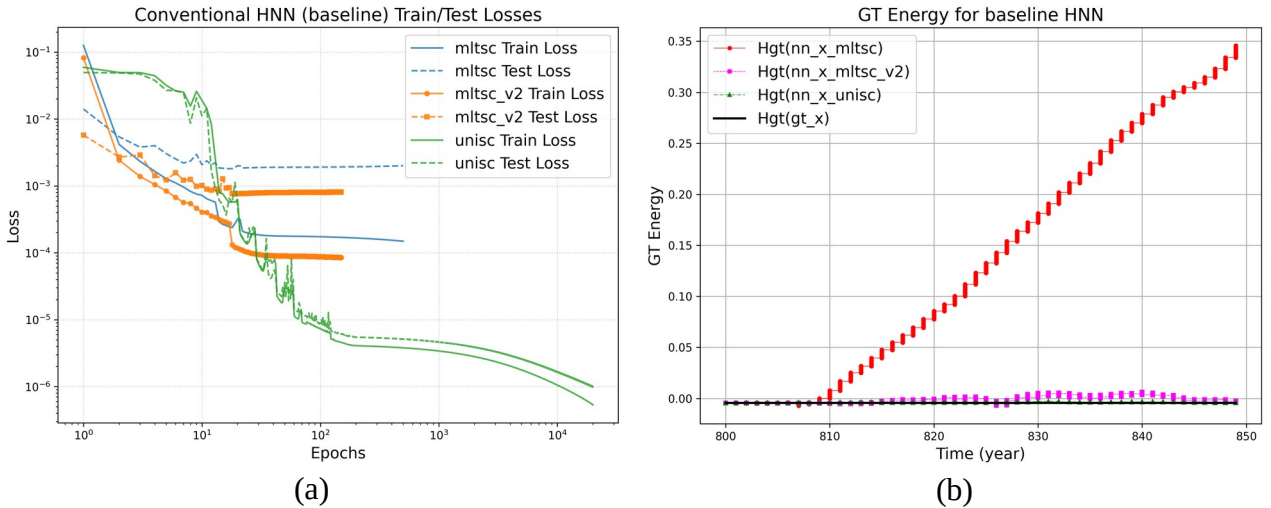


Figure 3: Comparison of baseline conventional HNN train/test losses and long-term Hamiltonian energy profiles over the 0–50 year orbit prediction interval across different scaling methodologies. (a) Evolution of training and testing loss profiles. (b) Ground truth Hamiltonian

function computed over true and predicted state variables (q, p). Here, 'unisc' denotes Global Uniscale normalization, 'mltsc' refers to Component-wise per-body scaling, and 'mltsc_v2' represents Vector-wise per-body scaling.

Initially, we sought to investigate the feasibility of utilizing a conventional Hamiltonian Neural Network (HNN) for the long-term orbital prediction of a multi-body system—specifically, a 10-body configuration of the Solar System—and to observe its baseline operational characteristics. In their foundational work, Greydanus et al. (2019) configured a network for a 2-body problem by stacking three linear layers with a hidden dimension of 200 neurons per layer. Drawing upon this design rationale and accounting for our 60-dimensional input state vector $(\hat{q}, \hat{p})$ for the 10-body system, we constructed a baseline HNN consisting of three linear layers with 240 hidden neurons per layer to conduct training and evaluation. For this optimization framework, we utilized a 1,000-year synthetic Solar System trajectory dataset, allocating the first 800 years as the training partition and reserving the remaining 200 years exclusively for testing loss measurement and long-term orbit prediction assessment.

To initiate the long-term orbit prediction assessment within the unseen domain, a single data point from the test partition was designated as the initial seed point for the iterative ODE integration. Using this baseline framework, we subsequently analyzed the impact of the three distinct scaling strategies—Global Uniscale, Component-wise Multiscale, and Vector-wise Multiscale V2—on the network's optimization path and resulting trajectory stability.

A primary challenge in modeling the Solar System is addressing the significant differences in physical magnitudes across celestial bodies, often characterized as a High Dynamic Range (HDR) problem. To systematically accommodate these varying physical scales, we applied three distinct scaling strategies to both the input state variables $(\hat{q}, \hat{p})$ and the corresponding target time-derivatives $(\dot{\hat{q}}, \dot{\hat{p}})$ across all data samples. These approaches—designated as Global Uniscale, Component-wise Multiscale, and Vector-wise Multiscale V2—are formally defined as follows:

$$\begin{aligned}\hat{X}_{b,i,j} &= \frac{X_{b,i,j}}{s}, \quad \text{with } s = \max_{b,i,j} |X_{b,i,j}| \quad (\text{uniscale}) \\ \hat{X}_{b,i,j} &= \frac{X_{b,i,j}}{s_{i,j}}, \quad \text{with } s_{i,j} = \max_b |X_{b,i,j}| \quad (\text{multiscale}) \\ \hat{X}_{b,i,j} &= \frac{X_{b,i,j}}{s_i}, \quad \text{with } s_i = \max_b \sqrt{X_{b,i,x}^2 + X_{b,i,y}^2 + X_{b,i,z}^2} \quad (\text{multiscale V2})\end{aligned}$$

where $i \in \{0, 1, 2, \dots, 9\}$ and $j \in \{x, y, z\}$

To evaluate the efficacy of conventional Hamiltonian Neural Networks (HNNs) under highly disparate physical scales, we analyzed the learning convergence, corresponding trajectory profiles, and long-term energy behaviors across the three normalization strategies (**Figure 3**, **Figure 4**, and **Figure 5**).

When employing Global Uniscale normalization, the system faces severe optimization stagnation due to the unmitigated High Dynamic Range (HDR) characteristics. To prevent immediate gradient explosion, an extremely conservative learning rate sequence (decaying to a minimum boundary of 1×10^{-6}) with a large batch size of 8,192 was employed. Under this constrained optimization layout, the baseline architecture demands approximately 20,000 epochs to establish a parameter space capable of maintaining a structurally bound state for a 50-year interval. However, extended training beyond this window yields no discernible performance improvements, identifying a practical orbit prediction boundary at roughly 50 years before numerical precession induces systemic disintegration.

Conversely, both the Component-wise Multiscale and Vector-wise Multiscale V2 frameworks resolve this inherent gradient stiffness by providing improved numerical conditioning across the feature coordinates. This mitigation forces an accelerated optimization path, triggering a steep drop in both training and testing loss bounds within fewer than 200 epochs. Yet, this rapid loss reduction fails to translate into physical validity over extended temporal horizons, revealing distinct modes of energetic failure. As mapped in the long-term energy tracking (**Figure 4**), the standard Component-wise Multiscale model undergoes a continuous, linear escalation in total Hamiltonian energy after a 10-year integration threshold, causing immediate structural decay. In contrast, the Vector-wise Multiscale V2 architecture yields bounded energy deviations that oscillate near the true Hamiltonian value; however, these geometric fluctuations still introduce localized instabilities over time. Ultimately, both scaling mechanics lead to a cascading destabilization event, culminating in a domino effect of unphysical inner planetary ejections within the localized domain (**Figure 5**).

Based on these findings, the Vector-wise Multiscale V2 was selected as our foundational normalization standard due to its superior learning convergence and relative energy control properties. This framework was subsequently applied to both the Plausible HNN and Block Plausible HNN architectures (as shown conceptually in **Figure 1(b)** and **(c)**) to structurally resolve the remaining trajectory drifts. To fully exploit these scaling characteristics, we first introduced the Plausible HNN (P-HNN), shifting away from generic black-box networks by enforcing an explicit Kinetic-Potential (T-V) energy separation and functional symmetry.

4.2 Performance Analysis of the Plausible HNN (P-HNN)

To establish a controlled evaluation, the initial Plausible HNN (P-HNN) configuration was designed to match the baseline dense model by employing three hidden layers with a scaling factor of $m=2$ ($l=3, m=2$), which corresponds to a hidden dimension of 270 units ($_{10}C_2 \times 3 \times 2 = 270$). Under the Vector-wise Multiscale V2 standard, this initial baseline P-HNN successfully maintains a gravitationally bound state for approximately 50 years (**Figure 6**), replicating the maximum lifespan achieved by the slow-converging Uniscale baseline but with a drastically accelerated optimization path (< 20 epochs; **Figure 7**).

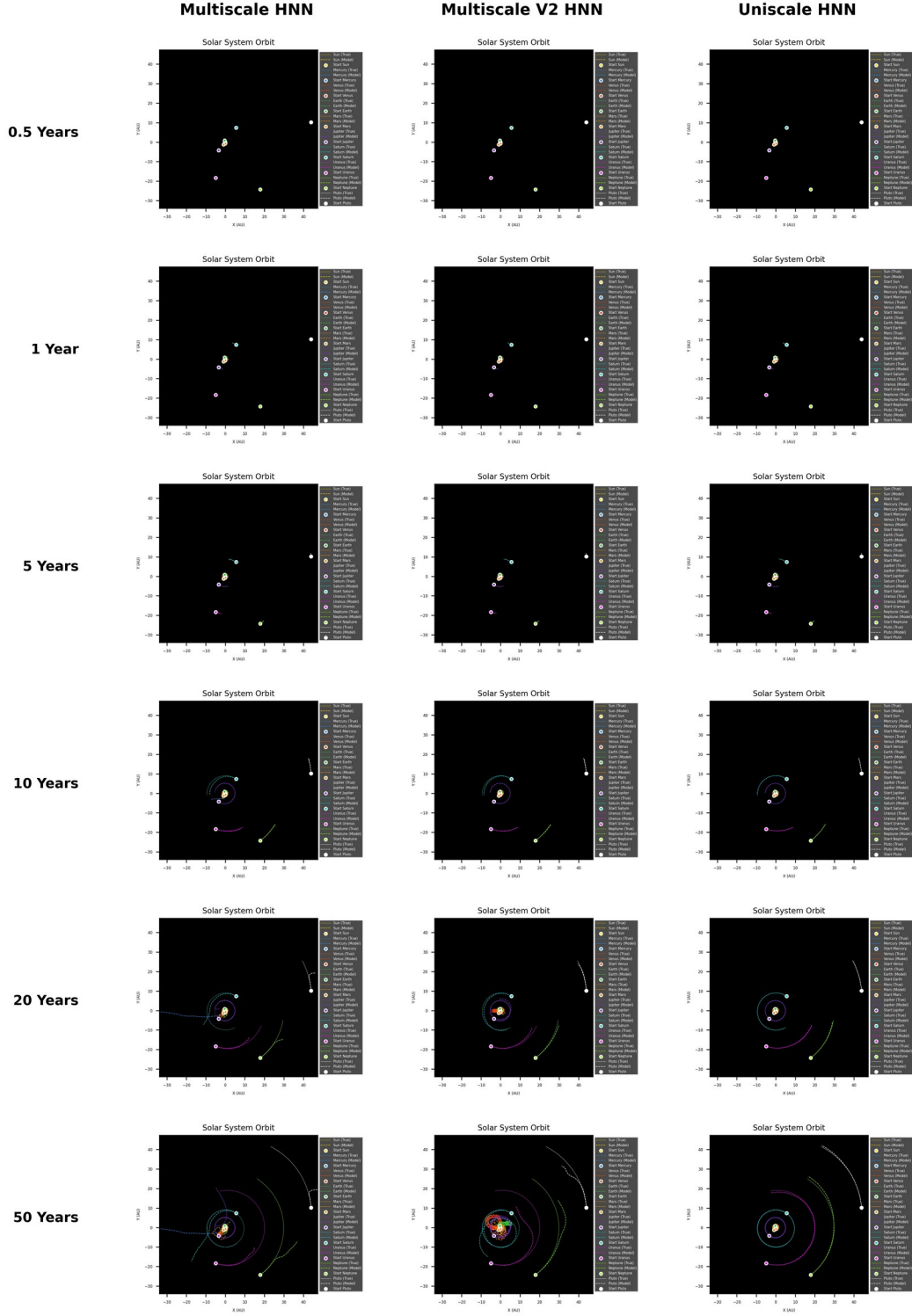


Figure 4: 50 years of orbit predictions for the Solar System with baseline HNNs. Macro-view of 10-body of the Solar system orbit predictions by feeding HNNs with the 1st point of test region dataset. Rows denote integration spans (0.5 to 50 years), and columns represent scaling formulations within the baseline dense HNN. The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

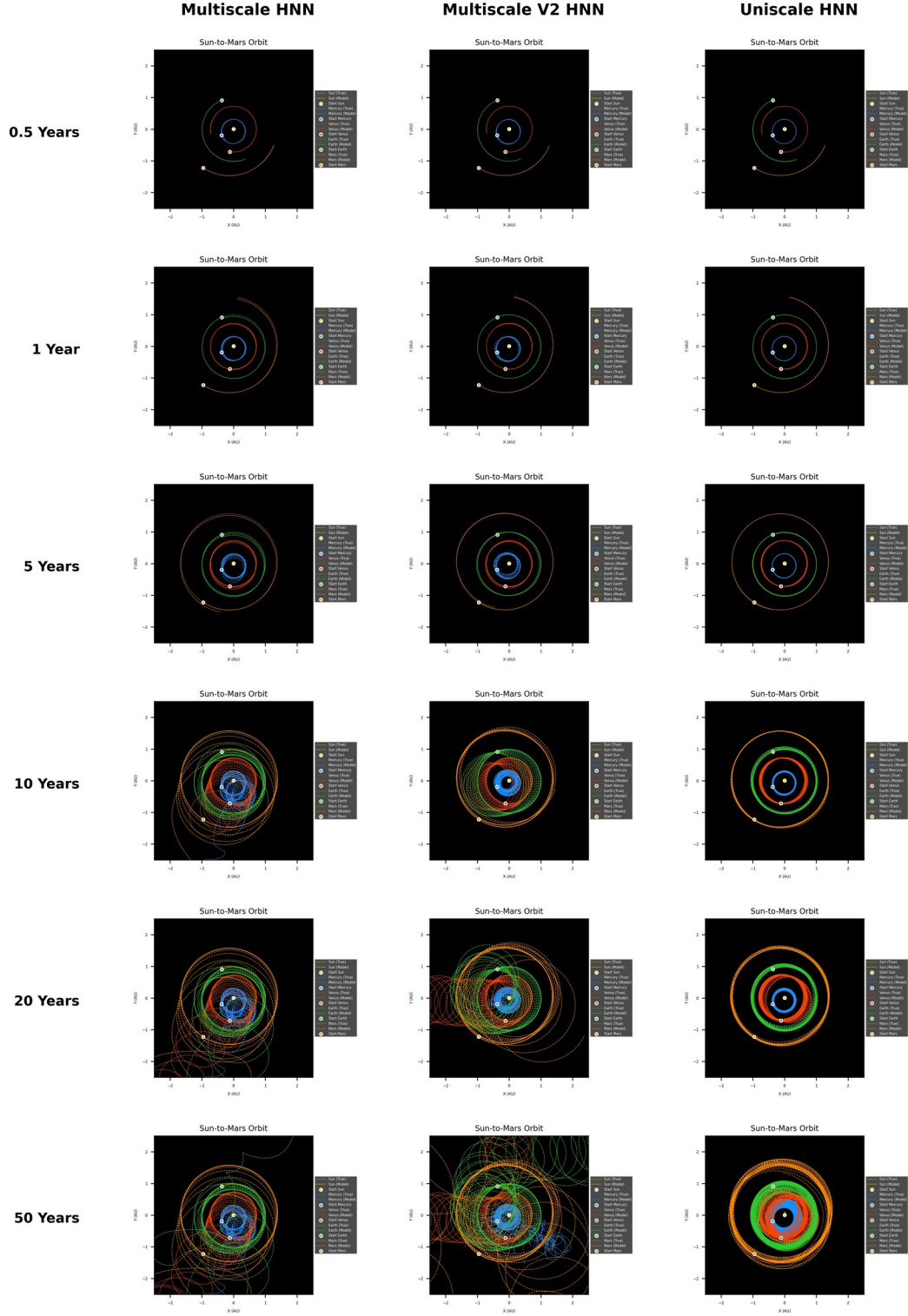


Figure 5: Predicted inner-planetary orbits detailed, highlighting rapid structural decay. Close-up view focusing on high-frequency inner planets (Mercury to Mars) under identical time horizons and scaling strategies as Figure 4. This demonstrates instantaneous trajectory ejections in 5 to 10 years for rapid-converging multiscale models. The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

P-HNN ($l=3, m=2$)

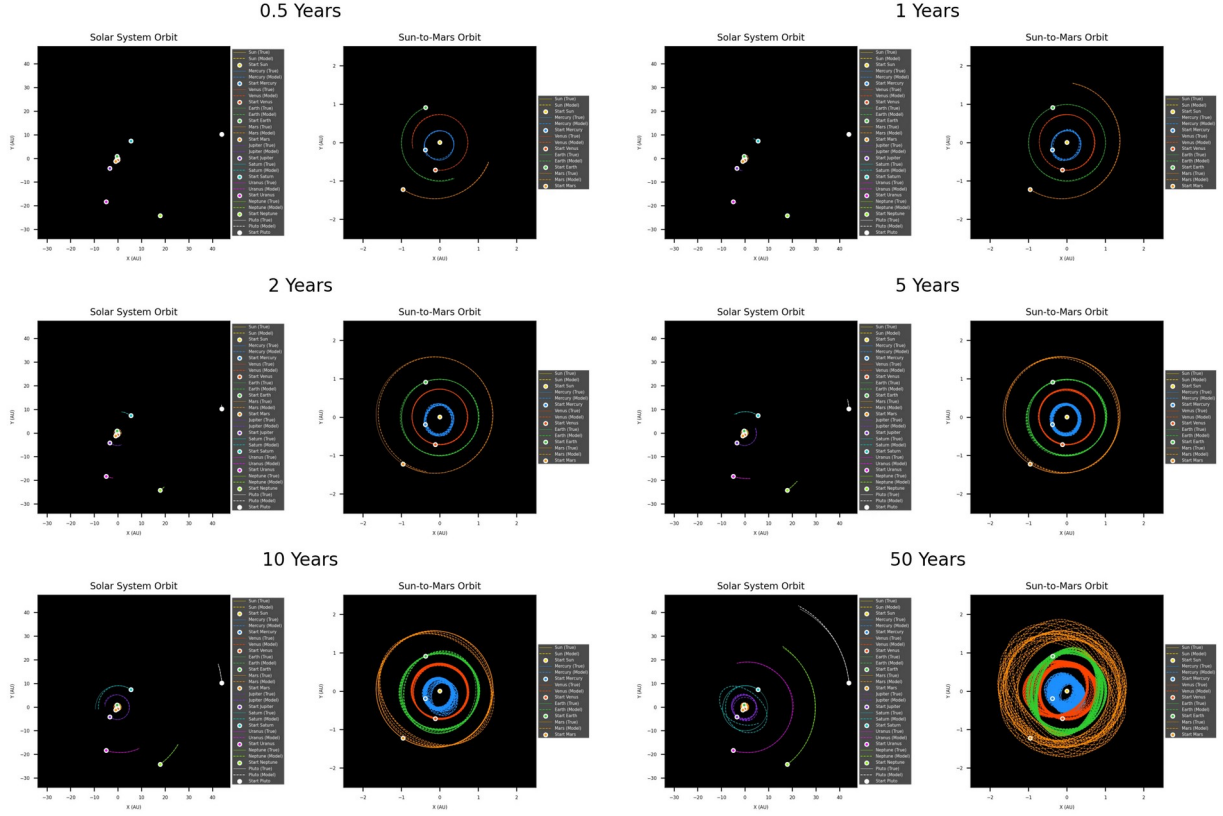


Figure 6: 50 years of orbit prediction by PlausibleHNN (P-HNN) for layers $m=3$ and factor $m=2$, which corresponds to 270 hiddens per layer. The model was trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

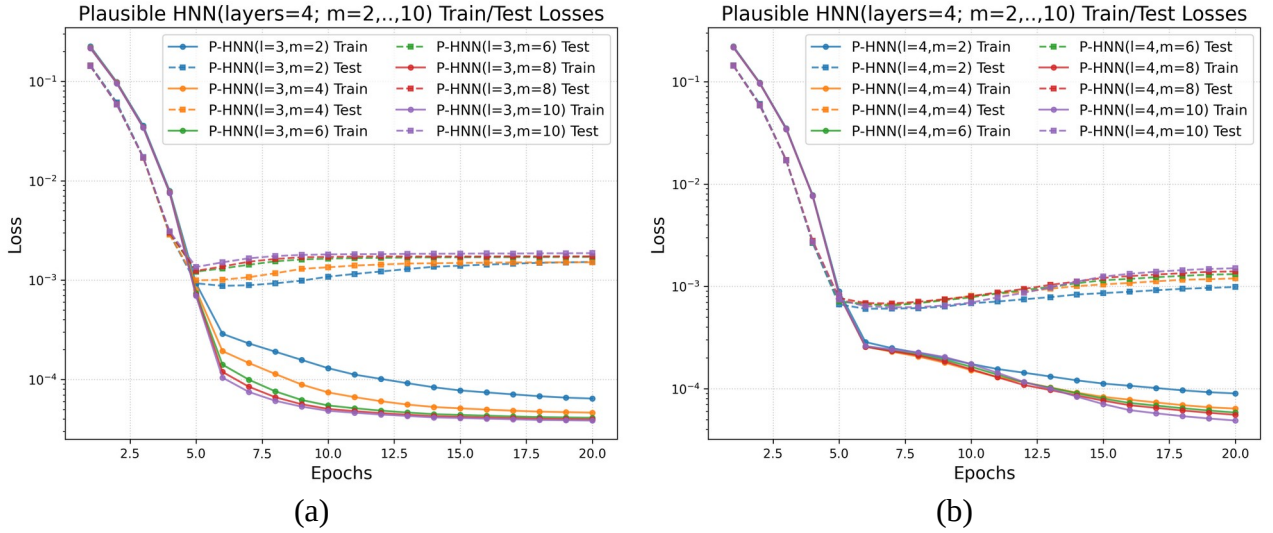


Figure 7: PlausibleHNN train/test losses: (a) layers $l=3$, factor $m=2, 4, \dots, 10$ and (b) layers $l=4$, factor $m=2, 4, \dots, 10$.

To improve the model's trajectory representation, we systematically expanded the network capacity by scaling the layer width parameter across $m \in \{4, 6, 8, 10\}$. While increasing the internal features generally showed a tendency to extend the orbital lifespan, the $l=3$ dense configuration encountered a strict operational ceiling and failed to sustain stable bounds beyond the 200-year horizon except for the $l=3$ and $m=8$ case (**Figure 8** and **Figure 9**).

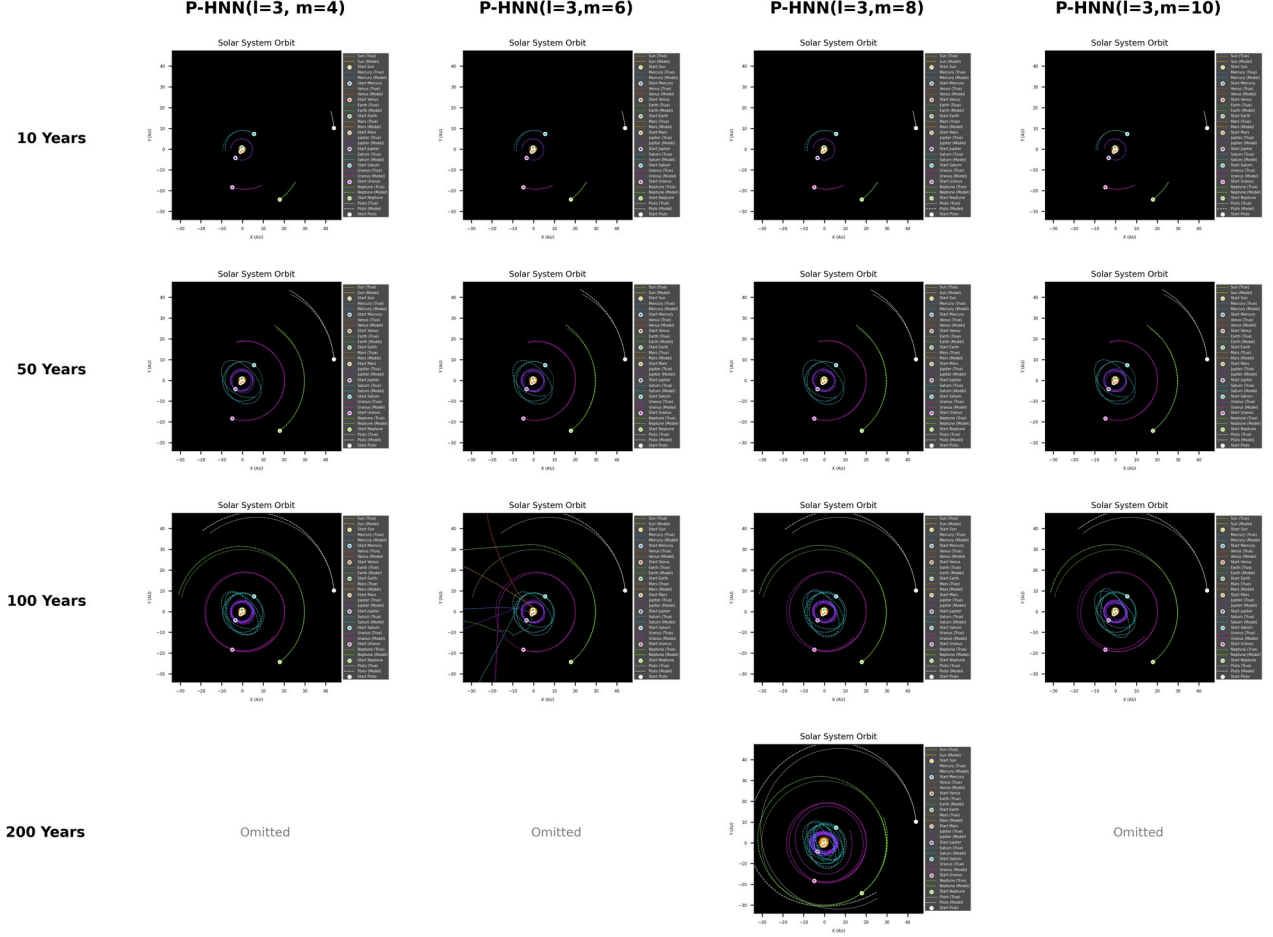


Figure 8: 200 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers $l=3$ and factor $m=4, 6, 8, 10$ by feeding P-HNNs with the 1st point of test region dataset. Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

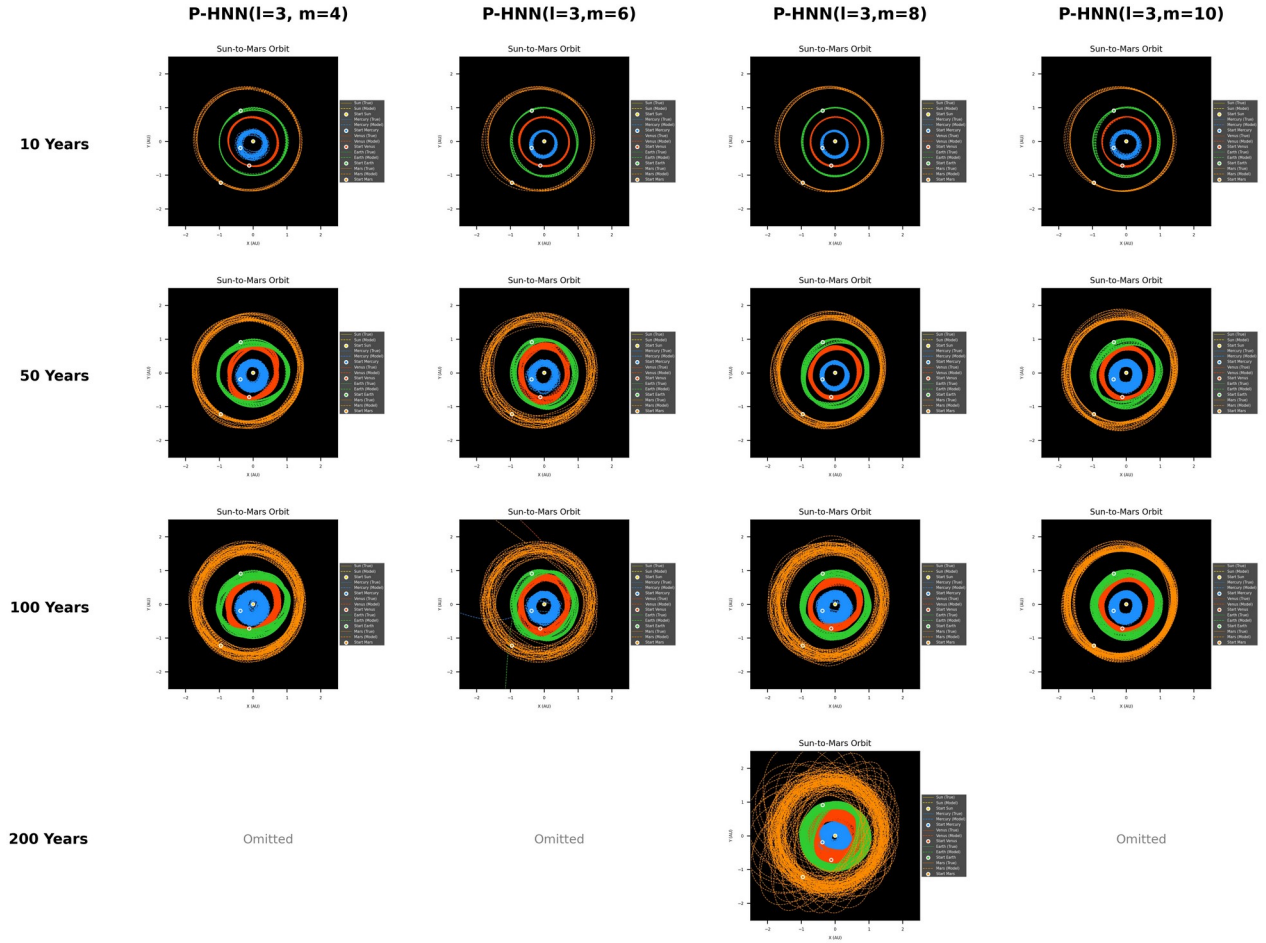


Figure 9: Detailed inner-planetary orbits predicted for 200 years with *PlausibleHNN (P-HNN)* for layers $l=3$ and factor $m=4, 6, 8, 10$ by feeding P-HNNs with the 1st point of test region dataset. Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

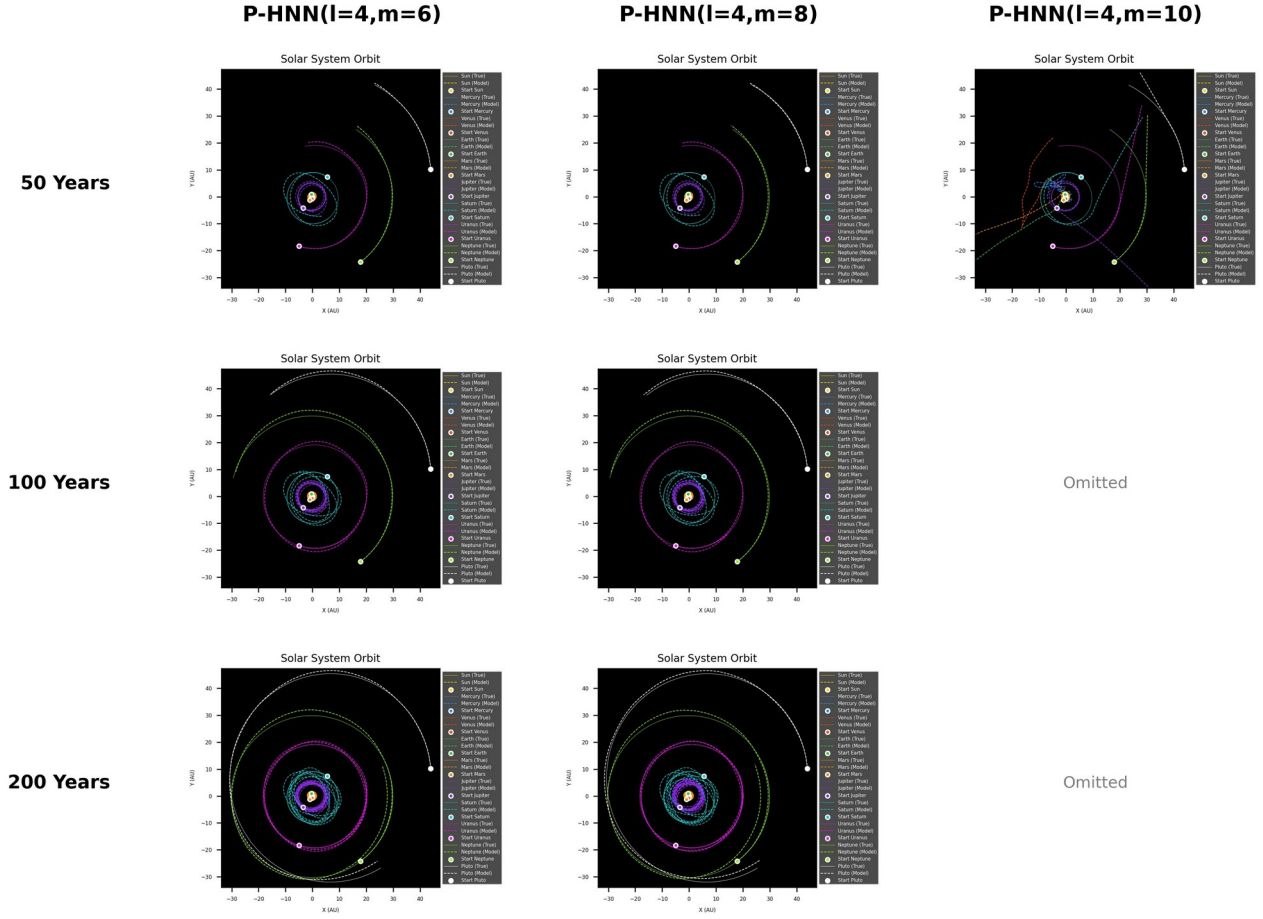


Figure 10: 200 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers $l=4$ and factor $m=6, 8, 10$ by feeding P-HNNs with the 1st point of test region dataset. Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

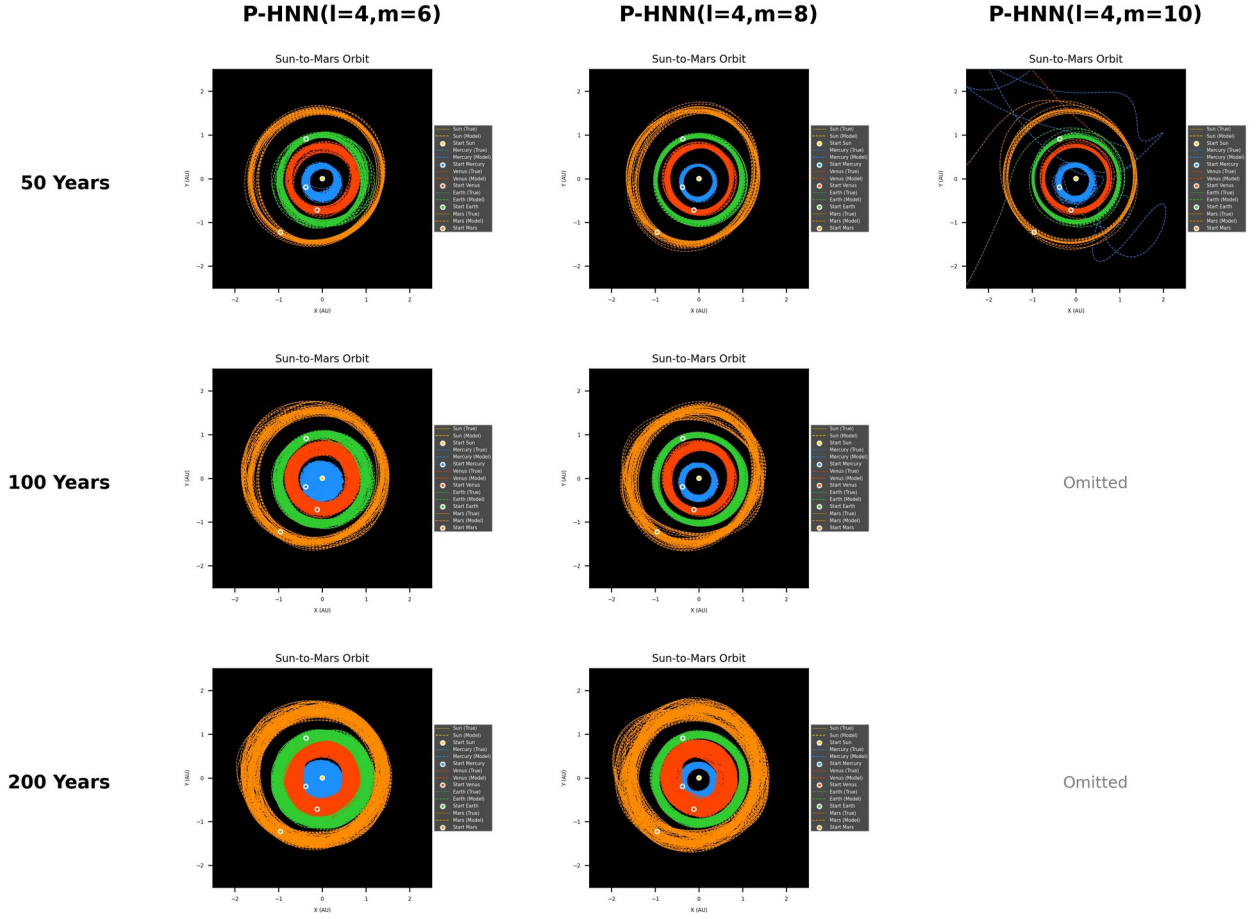


Figure 11: Detailed inner-planetary orbits predicted for 200 years with PlausibleHNN (P-HNN) for layers $l=4$ and factor $m=6, 8, 10$ by feeding P-HNNs with the 1st point of test region dataset. Apparently disintegrated orbits are not shown (omitted). The models were trained using the first 800 years data and evaluated for the remaining 200 years data from 1000 years dataset in $dt=0.0005$.

To break this geometric barrier, the structural depth was increased by adding an extra hidden layer ($l=4$). Empirically, the four-layer configuration with an enhanced feature width ($l=4, m=8$) successfully extended the system-level lifespan up to 200 years within the unseen test domain (**Figure 10** and **Figure 11**). Throughout this 200-year window, the planets experienced gradual, bounded numerical precession without catastrophic ejections. This explicit capacity expansion to ($l=4, m=8$), however, magnified the model's parametric footprint to approximately 57 MB.

The 200-year lifespan achievement of the dense P-HNN ($l=4, m=8$) of this 57 MB model size, represents a substantial advancement over the baseline configurations. However, it still falls short of capturing a complete orbital cycle of the furthest member in our 10-body setup: Pluto, which possesses an orbital period of approximately 248 years. For a learned Hamiltonian system to be considered dynamically complete and structurally stable over astronomical timescales, the integration window must surpass this 248-year threshold to ensure that the slowest high-dimensional phase space couplings are fully generalized rather than temporarily fitted.

To cover the prediction for at least one complete revolution of Pluto, we extended the dataset timeline up to 4,000 years using a 12th-order Yoshida integrator, allocating the first 3,200 years for training and the remaining 800 years for testing and evaluation. However, processing this dense

model over a high-fidelity 4,000-year dataset generated with a step size of $dt=0.0005$ (2,000 steps per year) created an intractable computational and memory bottleneck, as the raw files reached a massive 6 GB in total size (with $q, p, \dot{q}, \dot{p}$ occupying 1.5 GB each).

To resolve this operational overhead and facilitate rapid iterative validation, the 4,000-year temporal data was downsized by a factor of 20, sampling every 20th step to achieve a temporal resolution of $dt=0.01$ (100 steps per year). This downsampling effectively compressed the dataset footprint to approximately 402 MB (with $q, p, \dot{q}$, and $\dot{p}$ reduced to roughly 96 MB each). Crucially, verifying the dense P-HNN ($l=4, m=8$) architecture on this downsized temporal platform proved that the underlying physics remained invariant. The model successfully validated extended tracking capabilities, demonstrating that a specific deep configuration could sustain the bound state up to 400 years (**Figure 12** through **Figure 15**).

When the P-HNN models were trained with 3200 years and evaluated the remaining 800 years using the downsized 4000 years dataset with $dt=0.01$, we got P-HNN ($l=4, m=10$) of 88.6MB model size, as the best model this time, in view of overall orbit prediction performances in subsequent test partitions of 400 years window each: test partition 1 ranging from 3200 to 3600 years and the test partition 2 ranging from 3600 to 4000 years.

P-HNN Orbit Matrix (partition 1: 3200-3600 years)

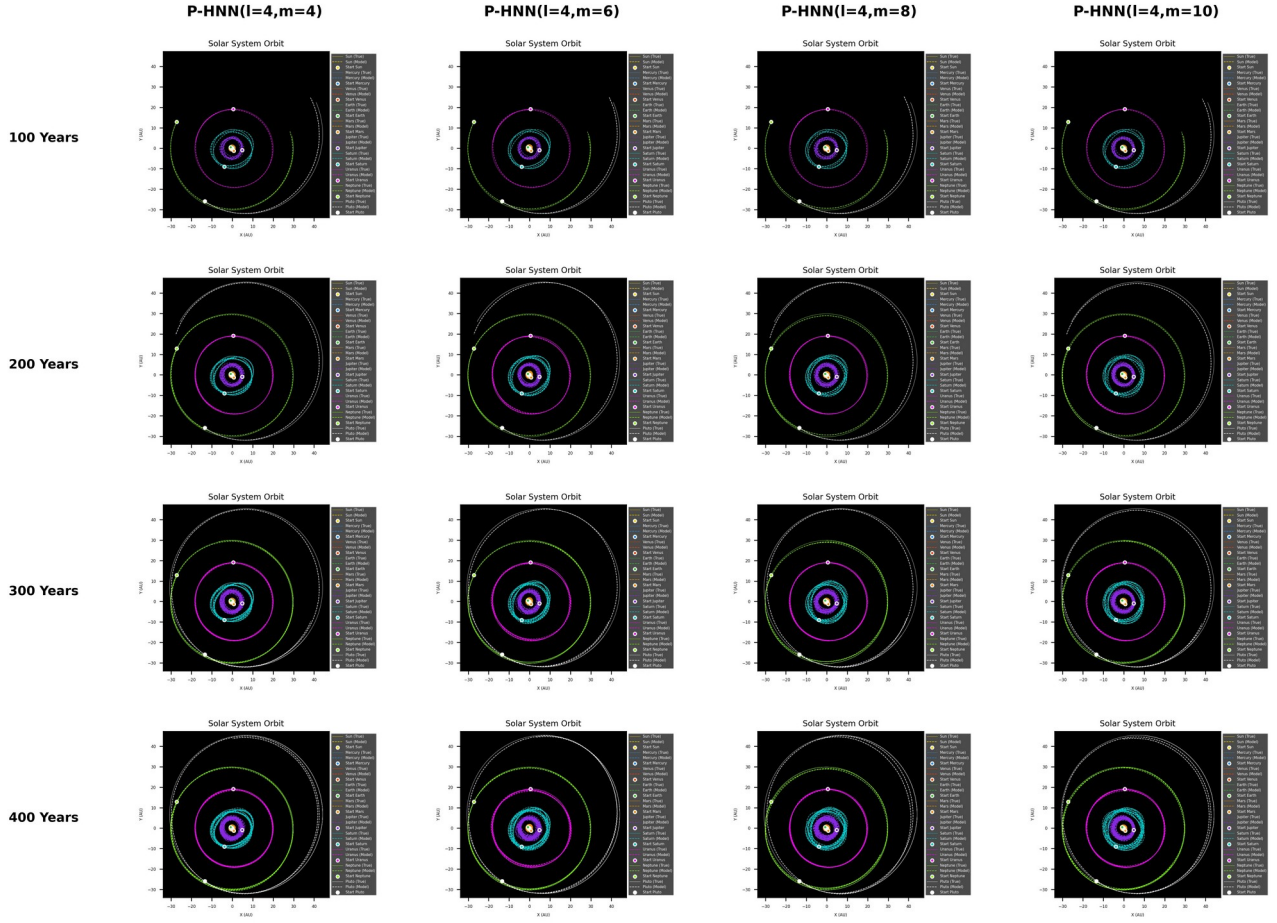


Figure 12: 400 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers $l=4$ and factor $m=4, 6, 8, 10$ by feeding P-HNNs with the 1st point of test partition 1 ranging from 3200 to 3600 years in downsized dataset with $dt=0.01$. The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in $dt=0.01$.

P-HNN Orbit Matrix (partition 1: 3200-3600 years)

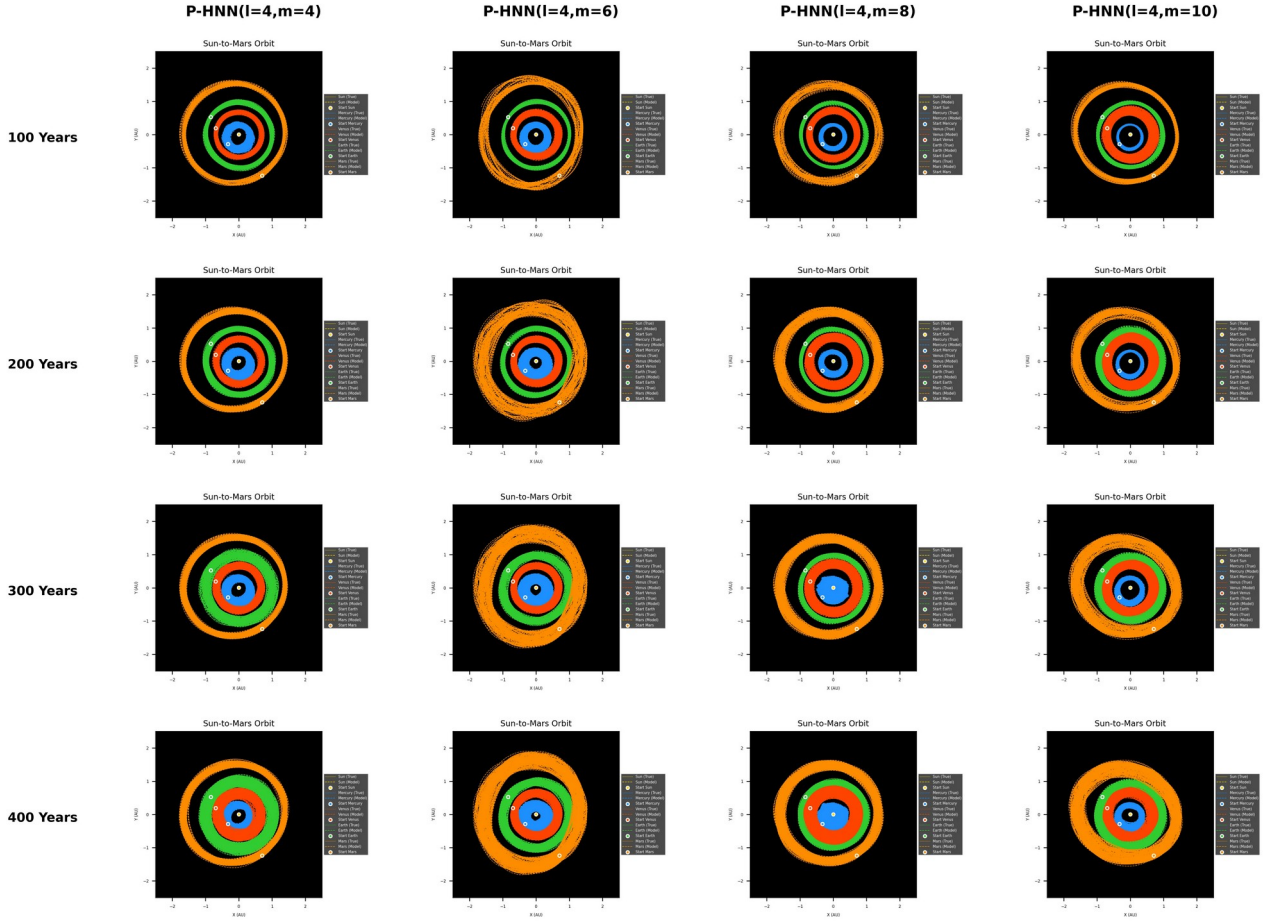


Figure 13: Detailed inner-planetary orbits predicted for 400 years with PlausibleHNN (P-HNN) for layers $l=4$ and factor $m=4, 6, 8, 10$ by feeding P-HNNs with the 1st point of test partition 1 ranging from 3200 to 3600 years. The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in $dt=0.01$.

P-HNN Orbit Matrix (partition 2: 3600-4000 years)

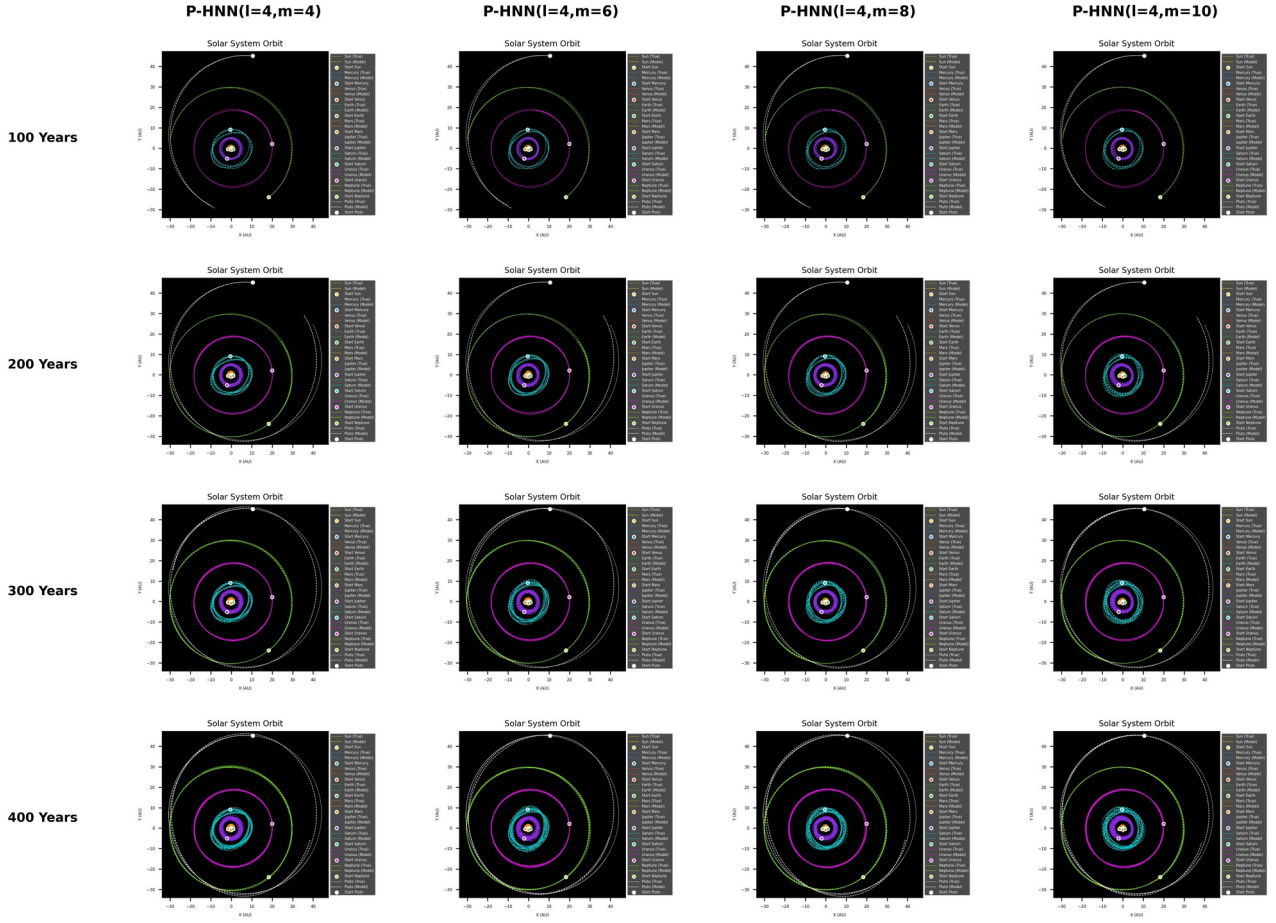


Figure 14: 400 years of orbit predictions for the Solar System with PlausibleHNN (P-HNN) for layers $l=4$ and factor $m=4, 6, 8, 10$ by feeding P-HNNs with the 1st point of test partition 2 ranging from 3600 to 4000 years in downsized dataset with $dt=0.01$. The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in $dt=0.01$.

P-HNN Orbit Matrix (partition 2: 3600-4000 years)

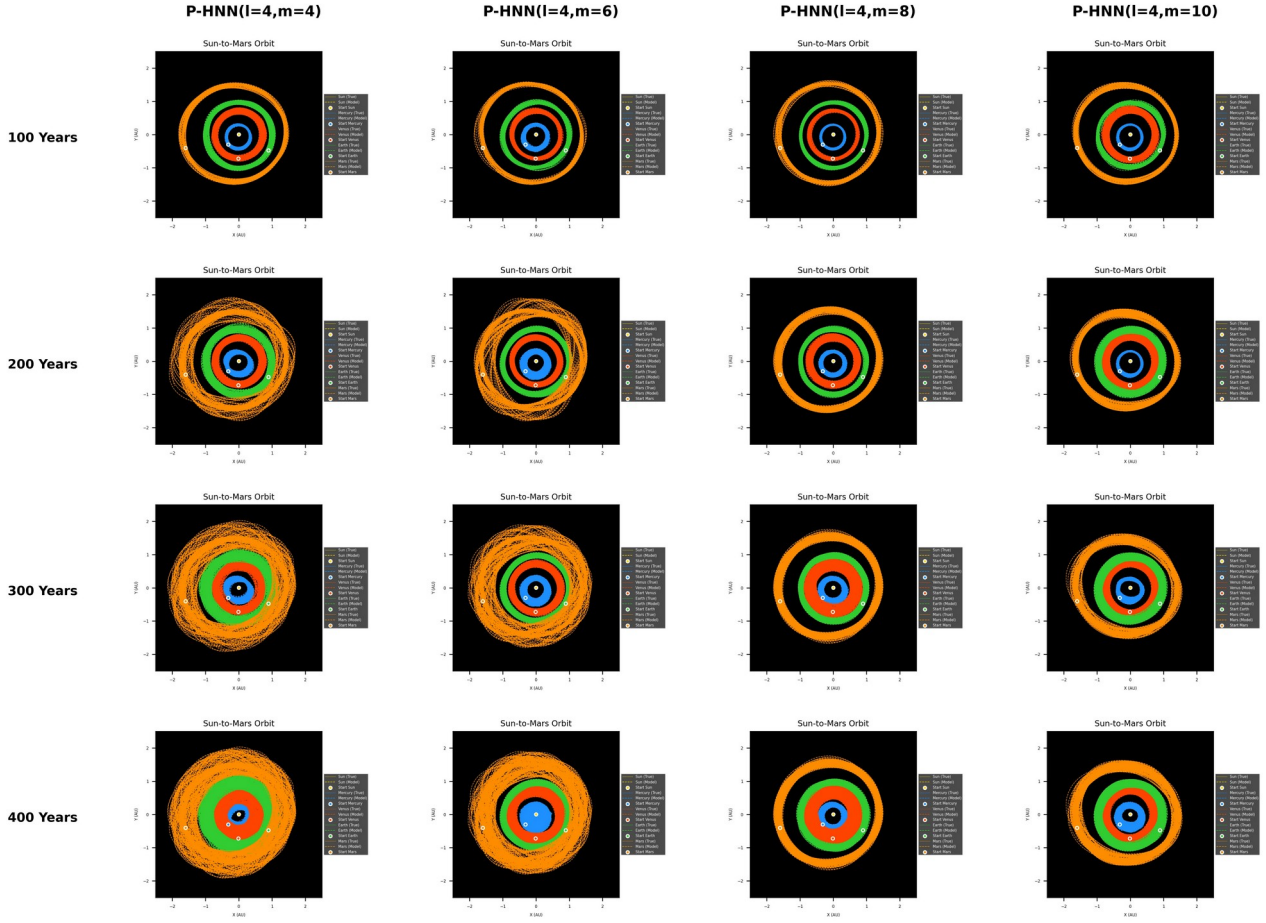


Figure 15: detailed inner-planetary orbits predicted for 400 years with PlausibleHNN (P-HNN) for layers $l=4$ and factor $m=4, 6, 8, 10$ by feeding P-HNNs with the 1st point of test partition 2 ranging from 3600 to 4000 years. The models were trained using the first 3200 years data and evaluated for the remaining 800 years data from 4000 years dataset in $dt=0.01$.

As illustrated in **Figure 16**, the energy profiles computed via both the ground truth Hamiltonian and the learned P-HNN Hamiltonian remain conserved with only minute fluctuations. This tight energetic alignment indicates that the overall multi-body system is successfully stabilized, preventing long-term structural collapse or unphysical planetary disintegration.

Nonetheless, the massive 88.7 MB memory footprint of this dense layout during extended inference motivated the final structural shift toward sparse partitioning. To surpass this boundary while simultaneously lifting the massive 88.7 MB computational load of the dense layout, we introduced the Block (Sparse) Plausible HNN (BLK-P-HNN) in the subsequent stage.

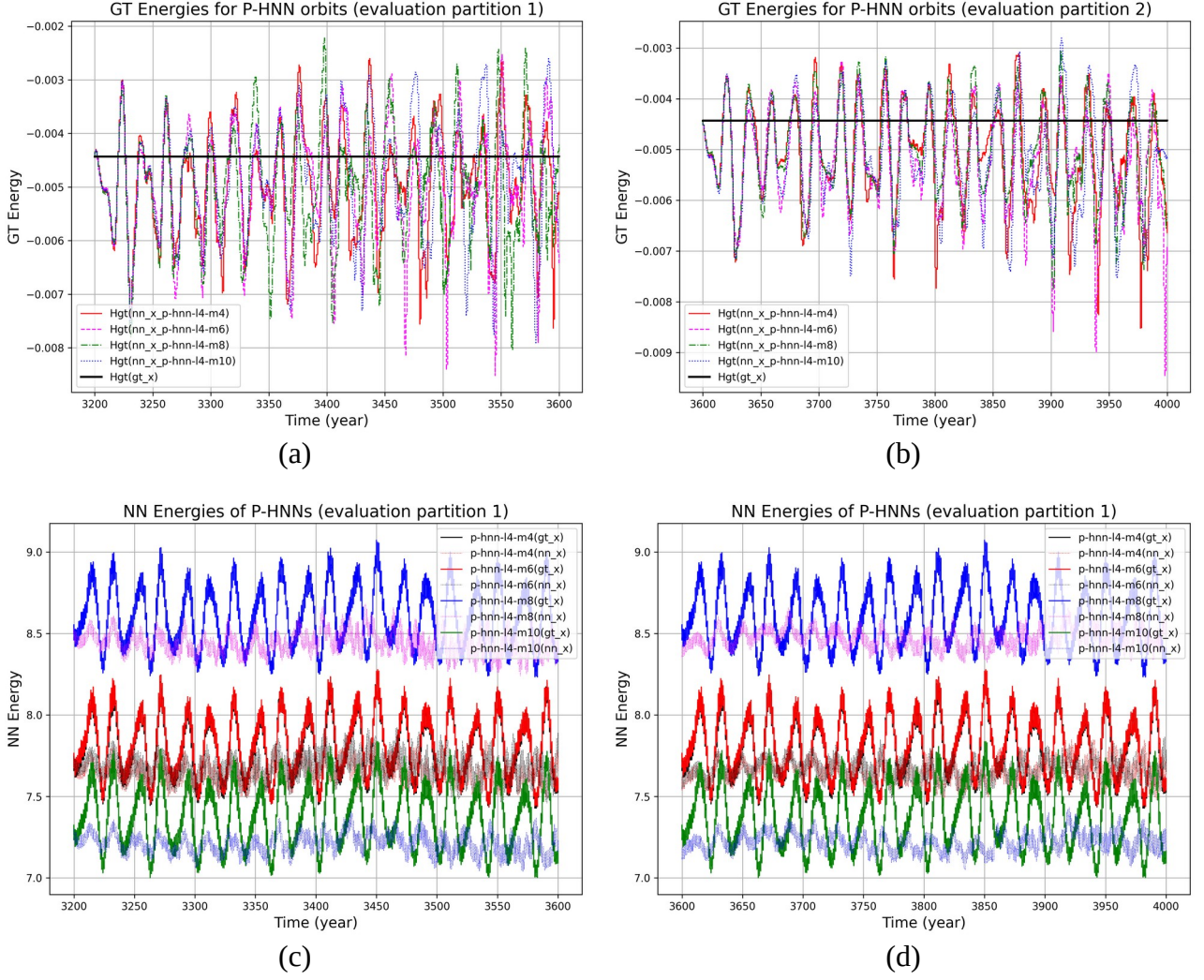


Figure 16: True and Network energies for true orbits and orbits predicted by PlausibleHNN (P-HNN) of layers $l=4$ and $m=4, \dots, 10$: (a) True energies for 3200-3600 years, (b) true energies for 3600-4000 years, (c) network energies for 3200-3600 years, and (d) network energies for 3600-4000 years.

4.3 Structural Optimization via Block (Sparse) Plausible HNN (BLK-P-HNN)

To address the computational bottleneck and memory footprint of the dense P-HNN without degrading its 400-year geometric stability, we evaluated the Block (Sparse) Plausible HNN (BLK-P-HNN) framework as shown in **Figure 1(c)** and **Figure 2**.

We consider only $l=4$ cases which have exhibited better performances in the Plausible HNNs than $l=3$ cases

Interestingly, the train/test loss profiles of P-HNN and BLK-P-HNN differ on the downsized 4000 years dataset as illustrated in **Figure 17**. In case of P-HNN, the best model for each l and m can be easily found by taking it at the epoch where the minimum of the test loss occurs. On the other hand, even after a sufficient initial decrease, the train/test losses of BLK-P-HNN tend to continuously

decline in a counterintuitive manner as training progresses. However, when inspecting the actual orbit prediction trajectories using weights from each subsequent epoch, a lower test loss did not necessarily translate to a superior model. Instead, the optimal model for each (l) and (m) within the BLK-P-HNN framework was found near the midpoint of the inspection window. This phenomenon indicates that the optimization landscape of BLK-P-HNN is implicitly ill-conditioned, suggesting that the optimal solution must be sought by intentionally truncating the iterations. This behavior closely resembles the L-curve criterion widely utilized in standard ill-posed inverse problems to determine the optimal regularization cutoff via iteration truncation [14, 15]. In this manner, the optimal model weight for BLK-P-HNN was obtained at $m=7$, in the aspect of the overall prediction performances throughout the evaluation partition 1 ranging from 3200 to 3600 years and the subsequently evaluation partition 2 ranging from 3600 to 4000 years. The orbit predicted with BLK-P-HNN ($l=4, m=7$) for 400 years, energies, and errors in positions/momenta is illustrated in **Figure 18** and **Figures 19**.

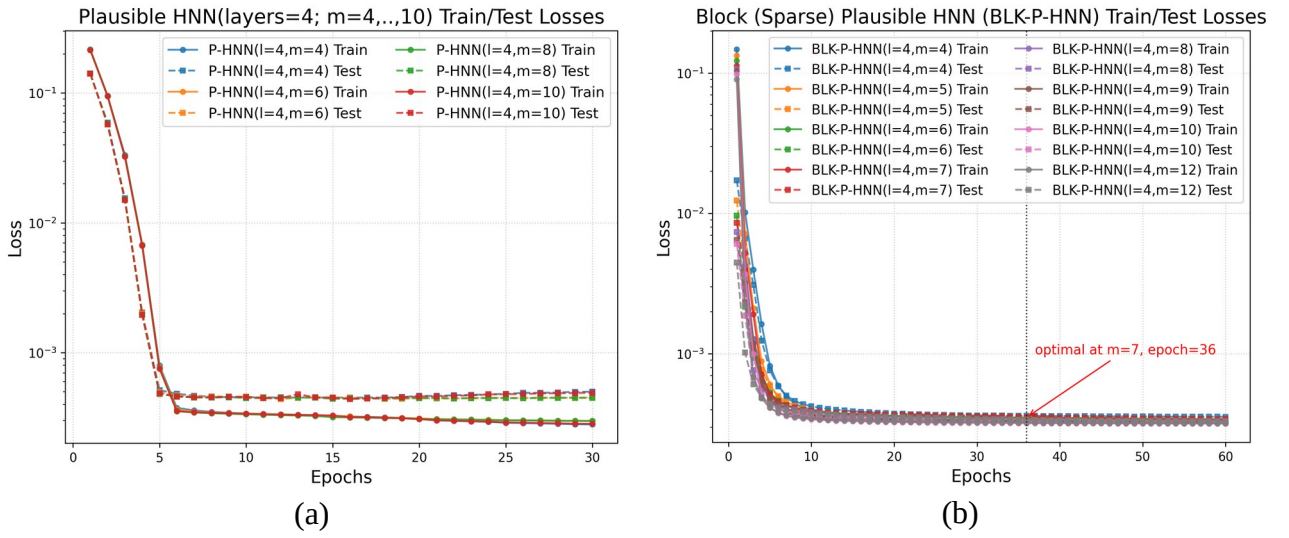


Figure 17: Train/test losses on the downsized 4000 years dataset with $dt=0.01$ for (a) Plausible HNN (P-HNN) and (b) Block Sparse Plausible HNN (BLK-P-HNN).

By utilizing the *Topology Adaptor*, which works as a pair-wise input linear adaptor, to isolate the position inputs into 45 independent sub-manifolds via batch matrix multiplications, the BLK-P-HNN effectively eliminates redundant cross-planet weight connections. This structural sparsity compresses the best model's total weight footprint from 88.6 MB at P-HNN($l=4, m=10$) down to 200 kB at BLK-P-HNN($l=4, m=7$), representing an approximate 400-fold parameter reduction. When subjected to the 400-year integration test window (partition 1 and 2 ranging from 3200 to 3600 years and from 3600 to 4000 years respectively), the sparse model still shows excellent generalization capabilities.

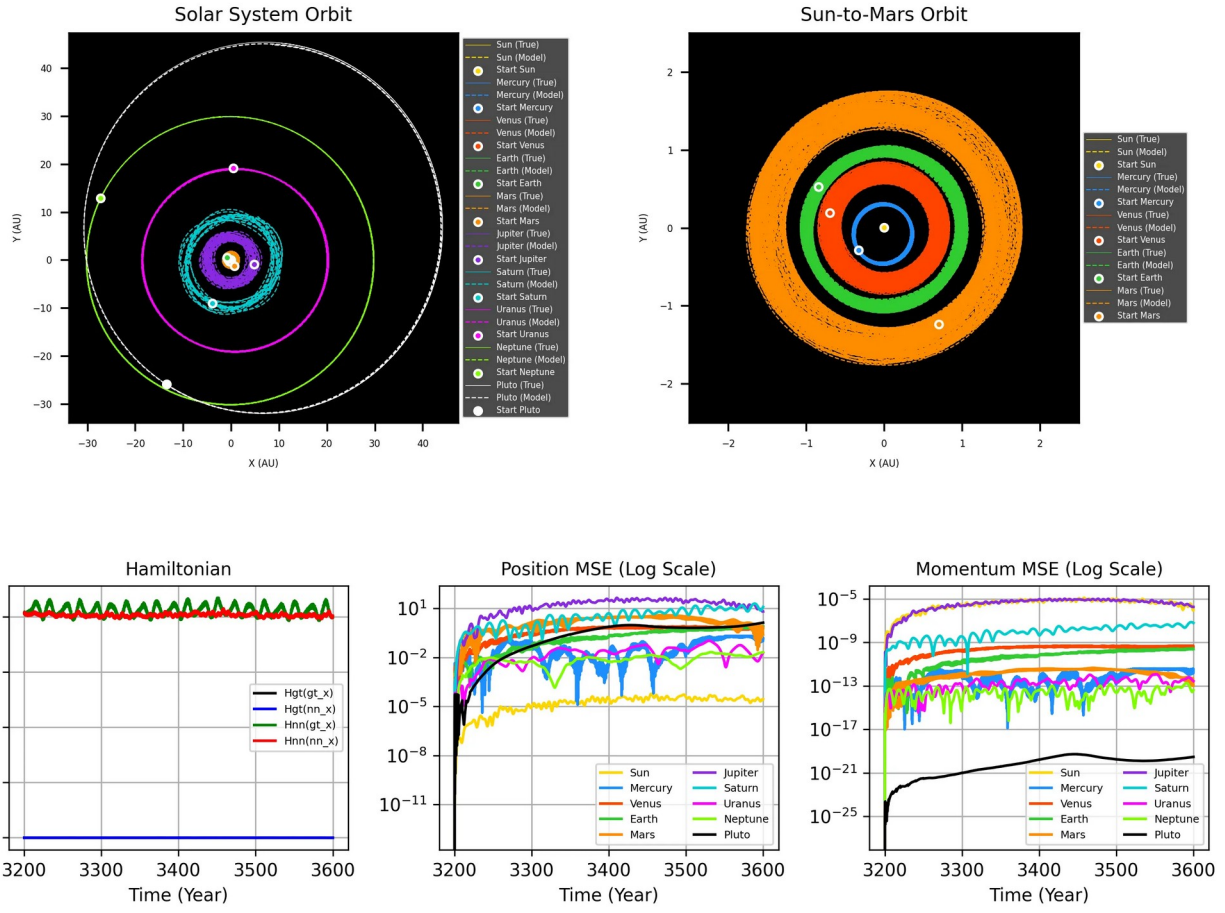


Figure 18: 400 years of orbit predictions and metrics for Block (Sparse) Plausible HNN (BLK-P-HNN) at the optimal configuration (layers $l=4$ and factor $m=7$) at evaluation partition 1 ranging from 3200 to 3600 years with $dt=0.01$.

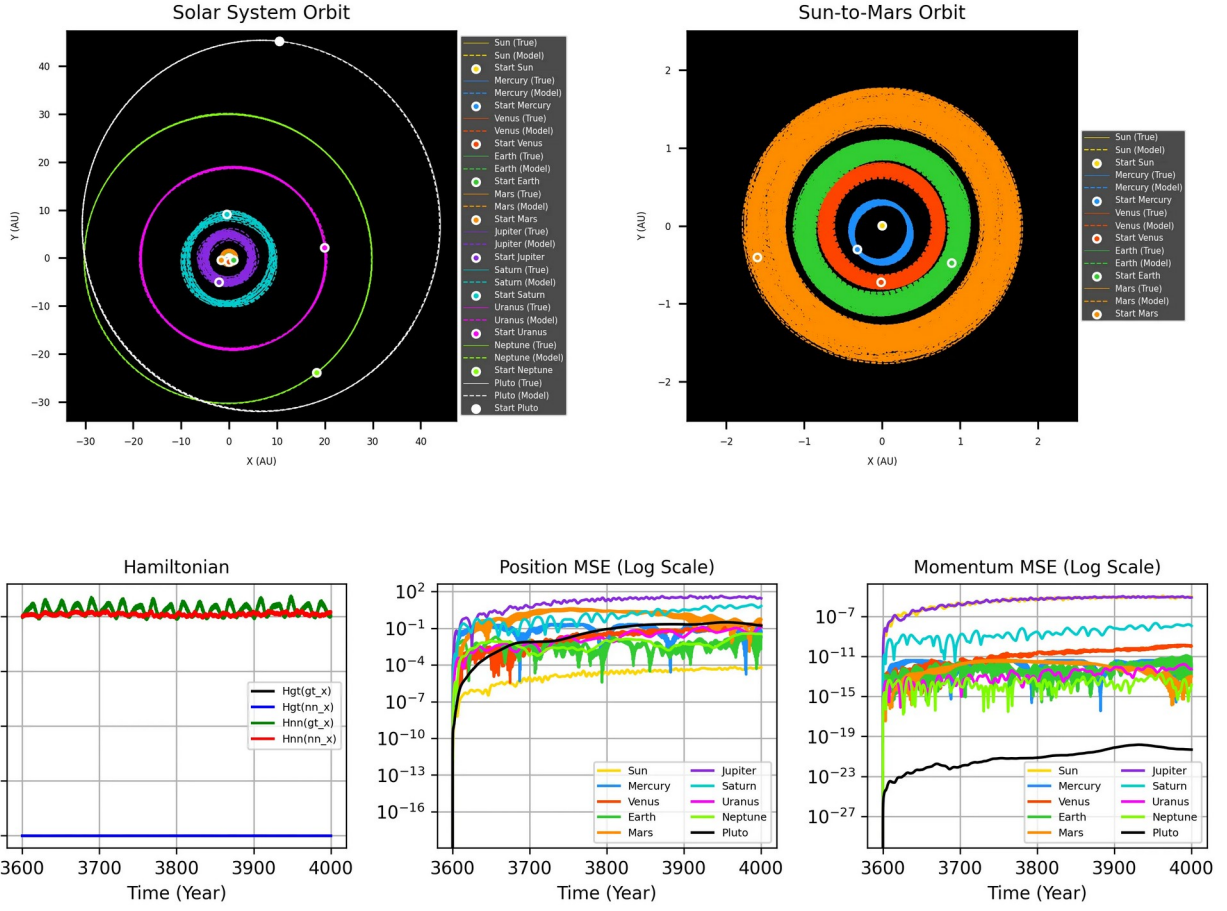


Figure 19: 400 years of orbit predictions and metrics for Block (Sparse) Plausible HNN (BLK-P-HNN) at the optimal configuration (layers $l=4$ and factor $m=7$) at evaluation partition 2 ranging from 3600 to 4000 years with $dt=0.01$.

4.4. Limitations of BLK-P-HNN

As the integration horizon expands, all neural-network-driven ordinary differential equation (ODE) integrators inevitably exhibit a distinct numerical precession phenomenon. Over centuries, this structural artifact manifests visually as a progressive broadening of planetary orbits into defined, band-like trajectories. This secular drift is an expected consequence of accumulated local truncation errors and infinitesimal gradient mismatches inherent to functional neural approximations over extended temporal scales.

Despite this inevitable band-widening effect, a stark contrast emerges regarding system-level stability, defined here as the sustainable orbital lifespan. While the conventional HNN governed by Vector-wise Multiscale V2 normalization suffers from catastrophic unphysical drift—leading to the rapid ejection of inner planets and complete structural collapse within a 10-year threshold—the proposed BLK-P-HNN maintains a strictly bound state for the entire 400-year integration duration. Under the BLK-P-HNN framework, the celestial bodies experience bounded numerical precession

but remain gravitationally locked within their realistic boundaries, never escaping the underlying Solar System manifold.

Nevertheless, a comparative analysis reveals that the average trajectory tracking performance of the dense P-HNN marginally outperforms the sparse BLK-P-HNN, as illustrated in Figures A1 and A2. When focusing exclusively on the XY-projections of the orbits, the BLK-P-HNN exhibits seemingly tighter, more localized spatial patterns. However, empirical inspection indicates that this superficial XY-localization is achieved at the expense of elevated orbital dispersions along the Z-axis. In contrast, the dense P-HNN demonstrates a more isotropic error distribution, where the trajectory dispersion in the vertical Z-direction does not exceed the corresponding error profiles observed within the horizontal X and Y dimensions.

4.5. Empirical Observations on BLK-P-HNN and Even Activations

Interestingly, in the architectures split into a kinetic net (T-net) and a potential net (V-net), we observed that the components of the attention weight vector become all 1/2 when trained. This empirical phenomenon seemingly indicates that the NN tends to interpret the ODE for the state variable in the scaled space as:

$$\dot{\hat{q}} = \hat{p}, \quad \dot{\hat{p}} = -\frac{\partial V}{\partial \hat{q}},$$

which structurally mirrors a canonical coordinate system. This intriguing behavior is consistently observed both in P-HNN and in BLK-P-HNN.

The positive potential energy that the V-net yields indicates that when we design the activation with an even function, the V-net tends to interpret the Solar system as a system similar to a complex system of coupled oscillators. The original design drive toward an even activation was the fact that the real potential energy is of the form of $V(\mathbf{r}_i, \mathbf{r}_j) = V(\sqrt{\mathbf{R}_{ij}^2})$, where $\mathbf{R}_{ij} = \mathbf{r}_i - \mathbf{r}_j$. In the approximation of V in terms of a power series expansion (i.e., Taylor series), only even terms in $\mathbf{R}_{ij}^{2k}$ ($k \geq 0$) survive due to spatial symmetry.

Since an NN can be viewed as a universal function approximator, we expected the V-net to learn this pair-wise symmetry upon the even activation when given some linear combinations of $\hat{\mathbf{q}}_i$ and $\hat{\mathbf{q}}_j$ via the *Topology Adaptor*. Crucially, we conjectured that in order to properly mimic the rich and diverse combinations of coefficients for these higher-order $O(2k)$ terms, a lateral amplification of neuronal numbers (expanding layer width) would be highly desirable and mathematically effective, rather than naively stacking deeper layers. This structural philosophy led us to the development of the P-HNN and subsequently to the BLK-P-HNN architecture. Ultimately, a long-term stable gravitational N-body system like our Solar system may be otherwise viewed as coupled complex oscillators under the potential of some constant value plus even terms in the relative distances—a phenomenological perspective that seems to align smoothly with the intrinsic representations learned by both the P-HNN and BLK-P-HNN frameworks.

Additionally, we examined the learned weights of the attention aggregation layer within the BLK-P-HNN. This attention mechanism was introduced with the intention of guiding the network to learn coefficients that correspond to the gravitational potential constants $Gm_i m_j$ within the Multiscale V2

scaled space ($\hat{\mathbf{q}}_i = \mathbf{q}_i / \max_{\text{sample}} \|\mathbf{q}_{\text{sample},i}\|$). When comparing the parameters of the trained model against the logarithmic values of the true, unscaled gravitational coefficients, $\ln(Gm_i m_j)$, a complete match was not achieved; however, a partial correlation was observed (**Figure 20**). Interestingly, the network automatically assigned the largest weights to pairs involving the Sun, and it captured limited similar patterns for interactions involving high-mass bodies such as Jupiter. This partial alignment suggests that the block sparse structure provides a baseline incentive for the model to explore physically meaningful parameter spaces, even though discrepancies remain.

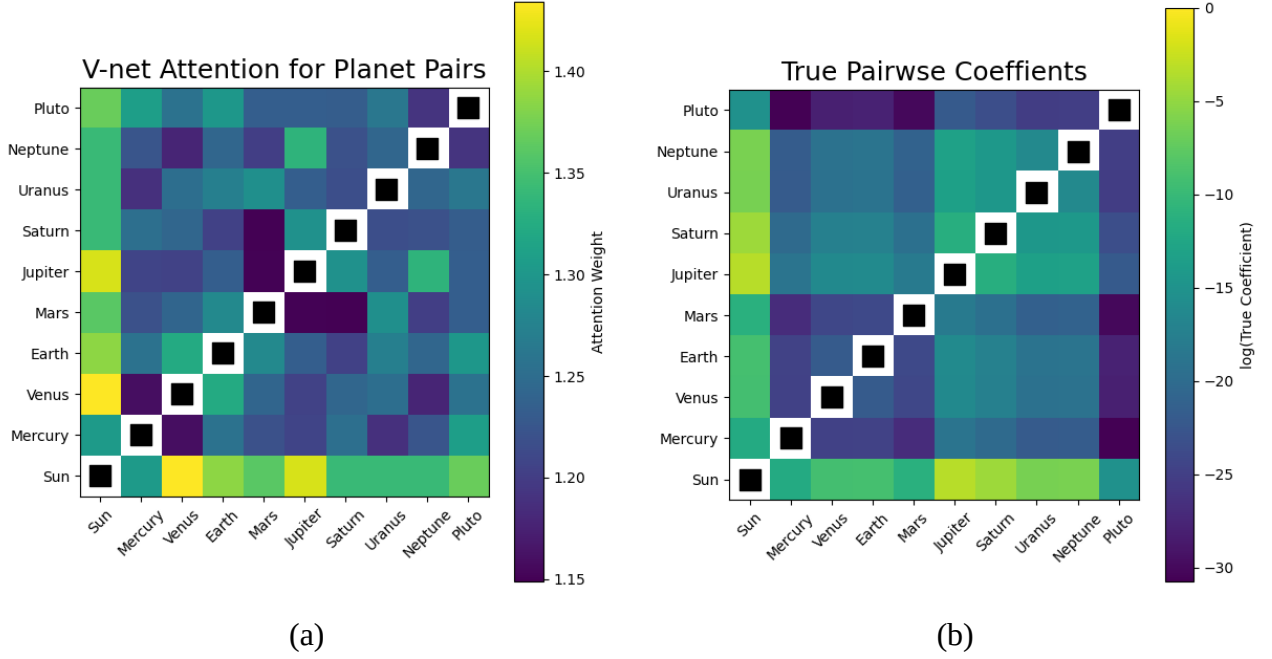


Figure 20: Comparison of V-net Attention and True Pair-wise Potential: (a) attention in V-net of BLK-P-HNN, (b) true $Gm_i m_j$ coefficients of the pair gravity potentials in the raw space.

5. Summary and Conclusions

In this work, we have systematically investigated the scaling boundaries of conventional black-box HNNs applied to high-dynamic-range, multi-body gravitational systems. While standard architectures exhibit severe gradient stiffness or long-term energetic failures leading to domino-effect planetary ejections, our integration of Vector-wise Multiscale V2 normalization and explicit (T-V) energy separation via the P-HNN extends the sustainable orbital lifespan past the critical 248-year orbital period of Pluto, effectively mitigating macro-scale system decay. To manage the parameter inflation and computational tractability constraints associated with long-term high-fidelity trajectories, we developed the Block (Sparse) Plausible Hamiltonian Neural Network (BLK-P-HNN). By leveraging a batch matrix multiplication-based Topology Adaptor to enforce structural sparsity, the BLK-P-HNN achieves a 400-fold parameter compression (from 88.6 MB to 200 kB). Empirical validation throughout a 400-year horizon reveals an informative performance trade-off: while the extreme structural compression introduces subtle localized orbital variations and an increased error dispersion concentrated along the vertical Z-axis compared to the dense model, the

sparse framework successfully maintains a gravitationally bound state without catastrophic orbital breakdown.

Beyond structural compression, empirical observations of the trained frameworks provide useful insights into neural representations of physical symmetries. By deploying even activation functions, the V-net was explicitly designed under the expectation that the network would learn the pairwise spatial symmetries found in the Taylor series expansions of gravitational potentials. Furthermore, based on our mathematical reasoning, we investigated the structural impact of width versus depth; while structural depth variations were constrained to a comparative evaluation between three- and four-layer configurations, scaling the features laterally demonstrated that expanding the hidden layer width—up to the boundary of over-parameterization—serves as an effective approach to capture the diverse combinations of higher-order potential coefficients. Ultimately, these findings demonstrate that incorporating physical inductive biases with structured sparsity offers a viable, informative pathway to balance extreme model compression with long-term macro-scale stability in complex multi-body celestial simulations.

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Appendices

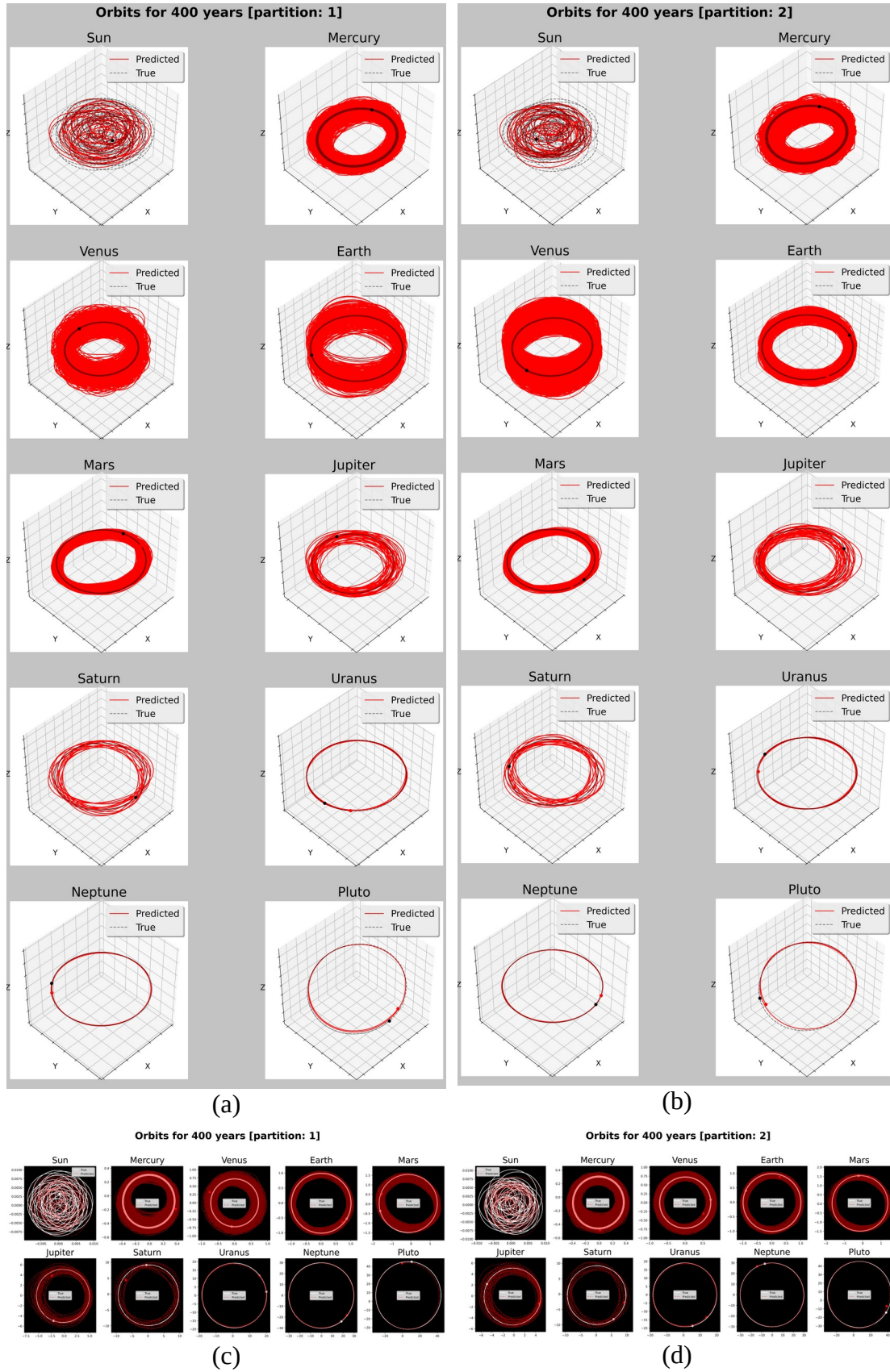


Figure A1: Orbit predictions per body with P-HNN($l=4$, $m=10$) in $dt=0.01$. (a) and (b) are 3D, (c) and (d) are 2D orbits, for partitons 1 and 2, each, where partition 1 ranging from 3200 to 3600 years and partition 2 ranging from 3600 to 4000 years.

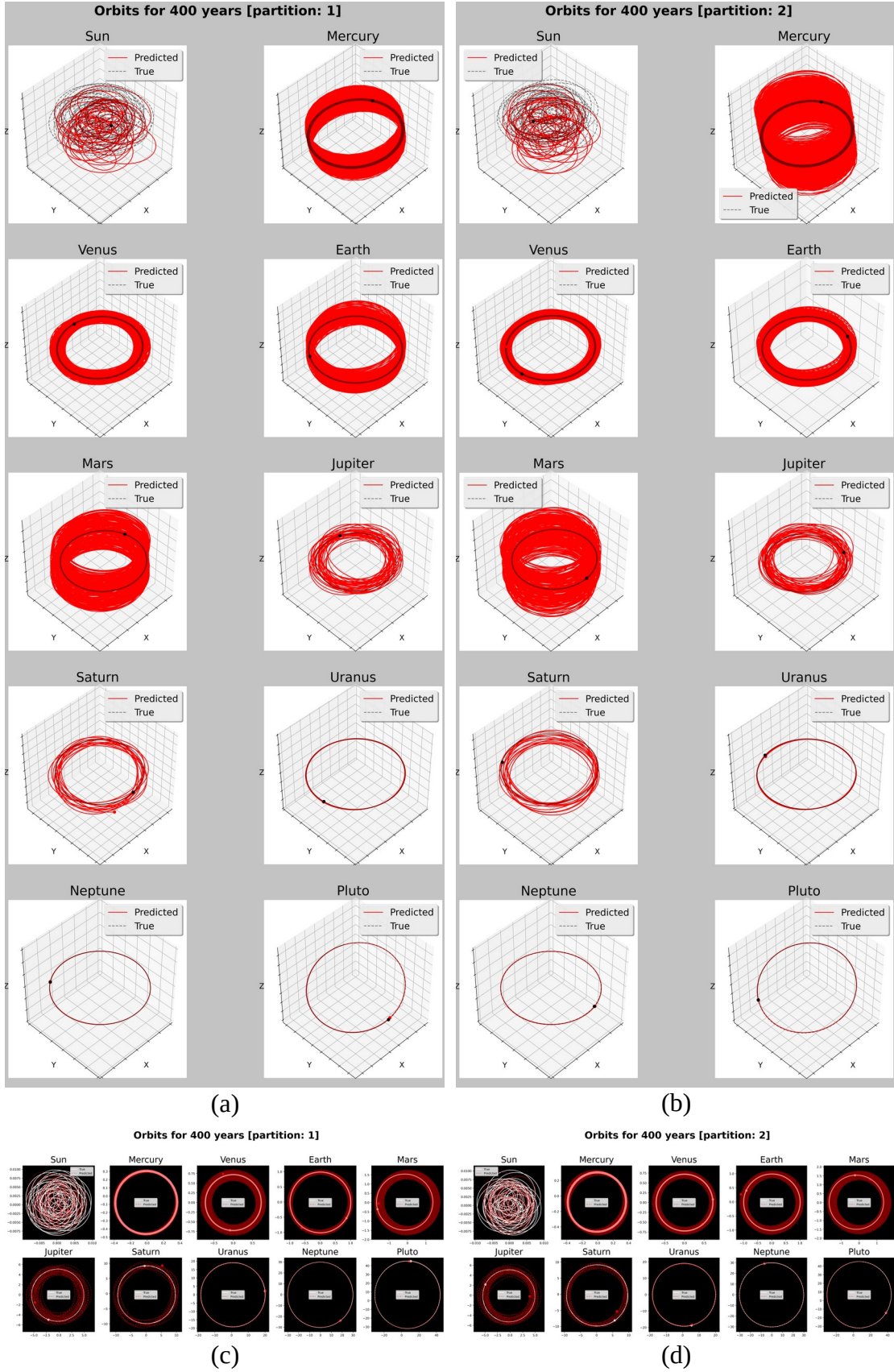


Figure A2: Orbit predictions per body with BLK-P-HNN($l=4$, $m=7$) in $dt=0.01$. (a) and (b) are 3D, (c) and (d) are 2D orbits, for partitions 1 and 2, each, where partition 1 ranging from 3200 to 3600 years and partition 2 ranging from 3600 to 4000 years.,

